

# AIR (Advice & Intelligence Relationship) System Architecture Design – V2.0

Version: 2.0 – BIAN-Aligned Architecture

Based on: System Architecture Design V1.0 + AIR BIAN-Aligned DDD v3

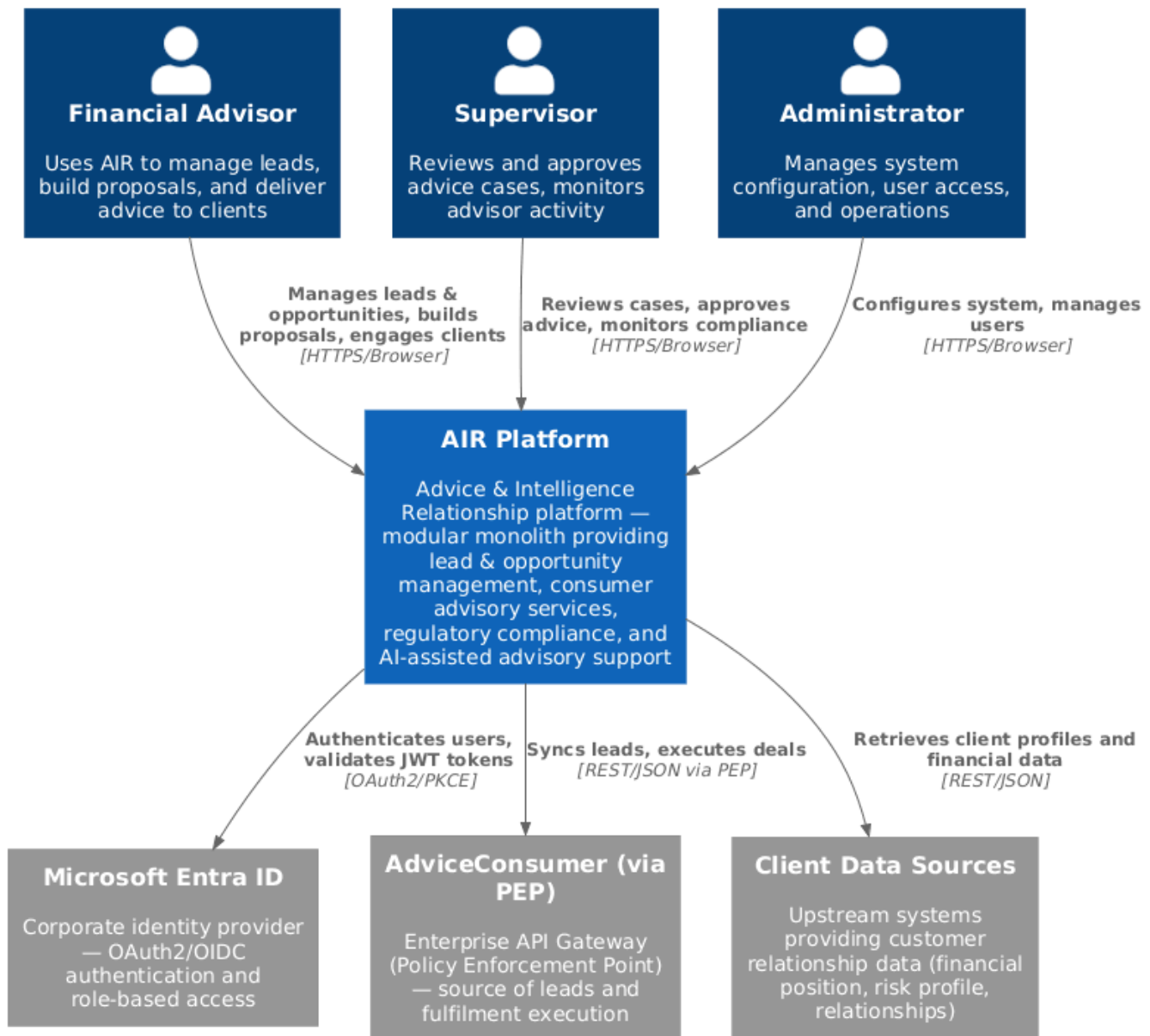
Date: May 2026

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## Level 1: System Context Diagram

Shows the AIR system in its environment, with external actors and systems.

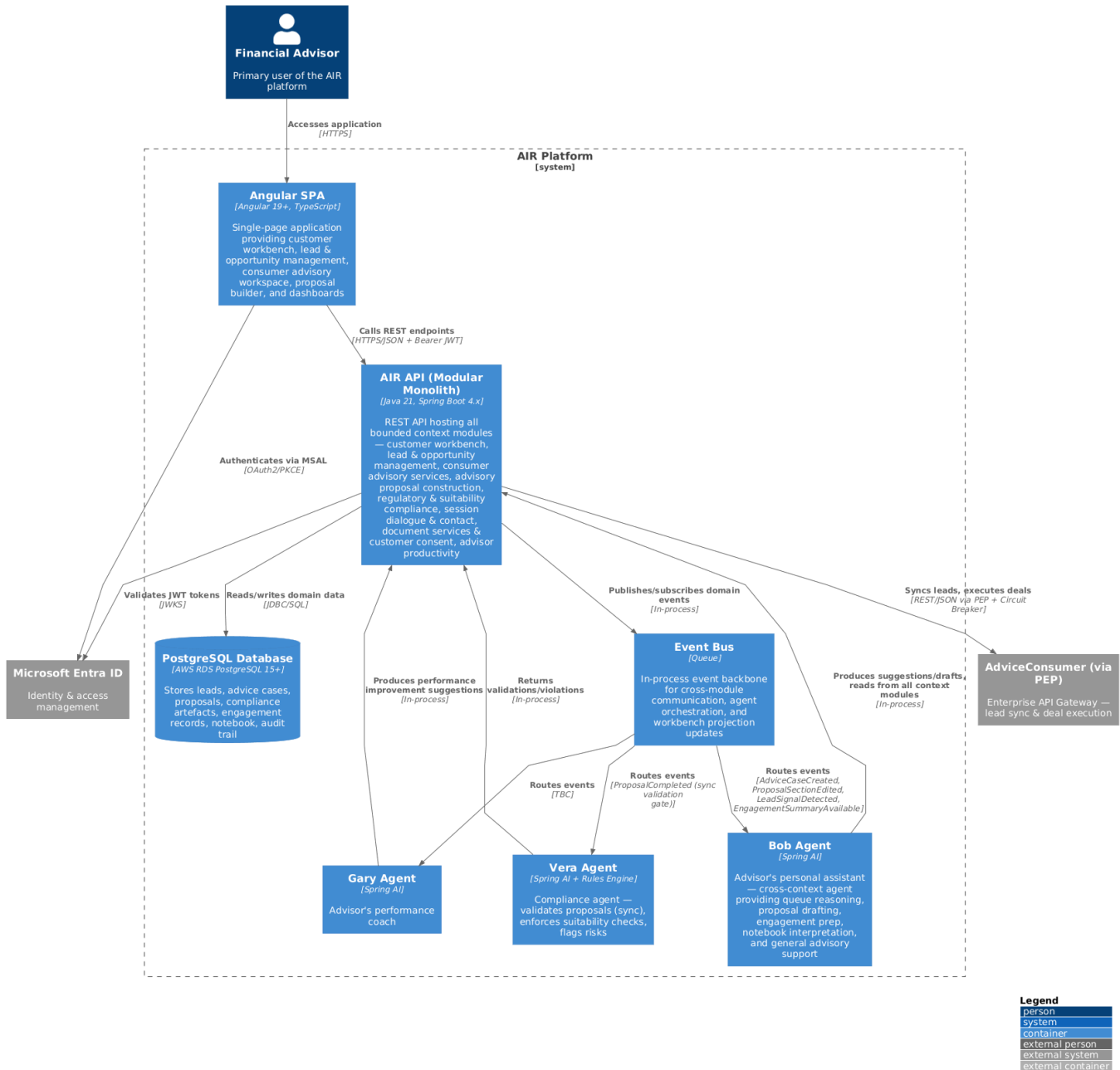
**AIR System Context Diagram**



## Level 2: Container Diagram

Shows the major deployable units and their interactions.

AIR Container Diagram

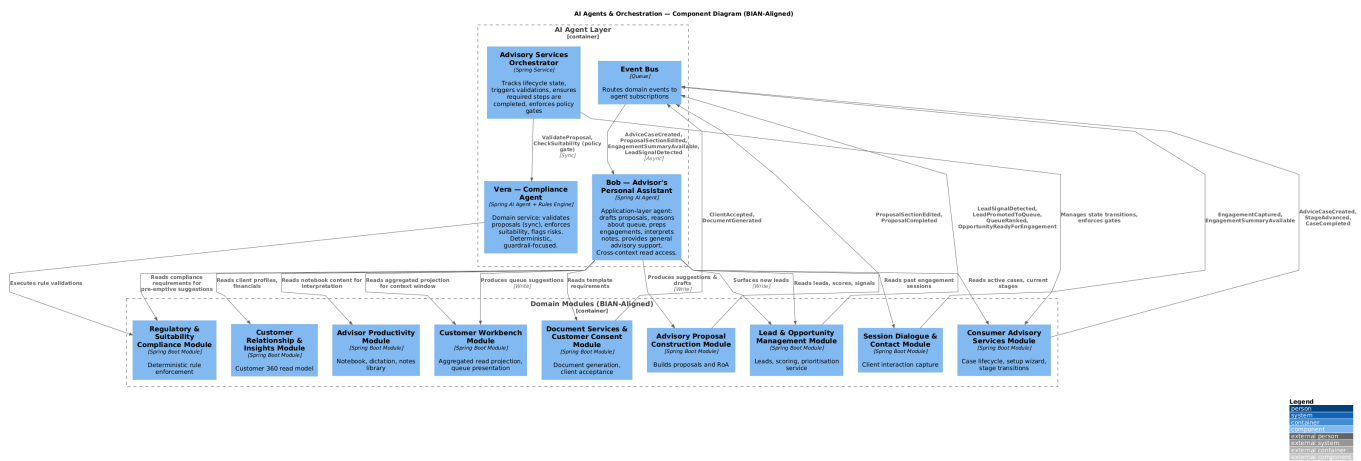


## Level 3: Component Diagram – AIR API (Modular Monolith)

Shows the internal module structure of the monolith, aligned to the BIAN v14 bounded contexts.

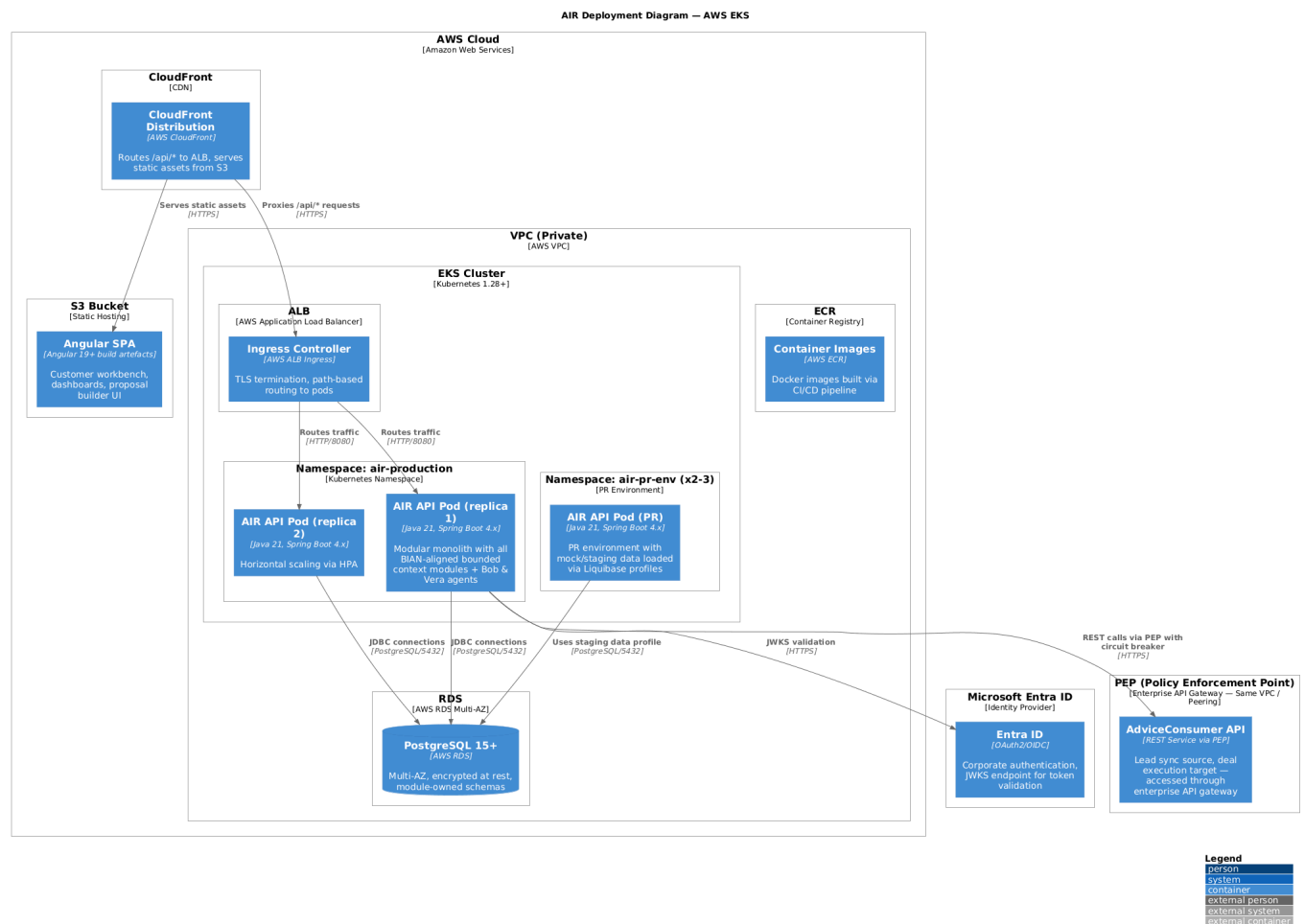
## Level 3: Component Diagram – AI Agents & Orchestration

Focused view of how Bob and Vera interact with the BIAN-aligned domain modules. Gary is deferred.



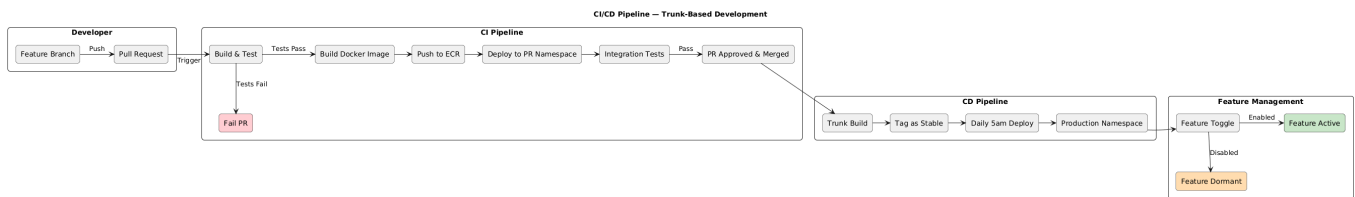
## Level 4: Deployment Diagram

Shows the physical deployment topology on AWS. Unchanged from V1.0 – deployment is infrastructure-level and independent of domain naming.



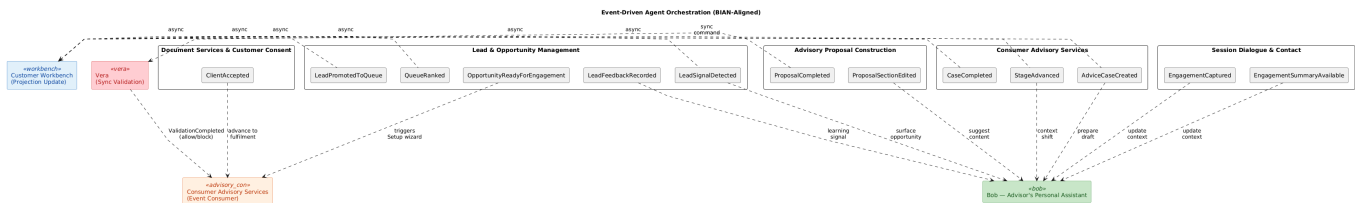
## CI/CD Pipeline Flow

Illustrates the trunk-based development and deployment pipeline. Unchanged from V1.0.



## Event Flow Diagram

Shows the event-driven orchestration pattern aligned to the BIAN v14 bounded contexts.



## Diagram Summary

| Level         | Diagram            | Purpose                                                                                     |
|---------------|--------------------|---------------------------------------------------------------------------------------------|
| C4 Level 1    | System Context     | Shows AIR in its ecosystem with users and external systems                                  |
| C4 Level 2    | Container          | Shows deployable units – SPA, API monolith, DB, Bob & Vera agents, event bus                |
| C4 Level 3    | Component (API)    | Shows internal BIAN-aligned bounded context modules within the monolith (9 active contexts) |
| C4 Level 3    | Component (Agents) | Shows Bob's cross-context read/write access and Vera's sync validation role                 |
| C4 Level 4    | Deployment         | Shows AWS infrastructure topology – EKS, RDS, CloudFront, S3                                |
| Supplementary | CI/CD Pipeline     | Shows trunk-based development and daily deployment flow                                     |
| Supplementary | Event Flow         | Shows event-driven orchestration with BIAN-aligned event catalogue                          |

## Key Architecture Characteristics

- **Modular Monolith:** All bounded contexts deployed as a single unit for speed, with explicit module boundaries for future decomposition
- **9 Active Bounded Contexts (BIAN-Aligned):** Customer Workbench, Lead & Opportunity Management, Customer Relationship & Insights, Consumer Advisory

Services, Advisory Proposal Construction, Regulatory & Suitability Compliance, Session Dialogue & Contact, Document Services & Customer Consent, Advisor Productivity

- **BIAN v14 Naming Convention:** All contexts use canonical BIAN service domain names where strong alignment exists; bank-specific extensions are clearly documented
- **Bob as Application-Layer Agent:** Cross-context personal assistant with read access to all modules and write access to Advisory Proposal Construction, Customer Workbench, and Lead & Opportunity Management
- **Vera as Domain Service:** Synchronous compliance validation within the Regulatory & Suitability Compliance module; async monitoring deferred
- **Gary Deferred:** Professional coach agent deferred to future phase (Servicing Activity Analysis context)
- **Prioritisation Service in Lead & Opportunity Management:** The Next-Best-Action ranking algorithm lives as a domain service within the Lead & Opportunity module, publishing ranked results to the Customer Workbench projection
- **Opportunity Setup Wizard in Consumer Advisory Services:** Triggered by `OpportunityReadyForEngagement` event from Lead & Opportunity Management, owned by Consumer Advisory Services
- **Event-Driven Backbone:** 13 domain events orchestrate cross-module communication and agent behaviour
- **Policy Gates:** Critical state transitions (e.g. `CompleteProposal`) go through synchronous Vera validation
- **AI Cannot Change State:** Agents suggest, flag, and advise – humans approve and transition
- **Feature Toggles:** Decouple code deployment from business release
- **PR Environments:** Ephemeral namespaces with mock data for rapid feedback
- **Module-Owned Data:** Each module manages its own schema via Liquibase

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## Alignment to BIAN-Aligned DDD v3

| Bounded Context (BIAN-Aligned) | Module                             | BIAN Service Domain(s)               | Classification | Notes                                                         |
|--------------------------------|------------------------------------|--------------------------------------|----------------|---------------------------------------------------------------|
| Customer Workbench             | <code>workbenchModule</code>       | Customer Workbench, Point of Service | Core           | Owns queue presentation, scorecard, aggregate read projection |
| Lead & Opportunity Management  | <code>leadOpportunityModule</code> | Lead and Opportunity Management,     | Core           | Owns lead scoring, pipeline,                                  |

| Bounded Context (BIAN-Aligned)       | Module                 | BIAN Service Domain(s)                                                                     | Classification    | Notes                                                         |
|--------------------------------------|------------------------|--------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------|
|                                      |                        | Customer Behavior Insights, Product Matching                                               |                   | prioritise service, feedback                                  |
| Customer Relationship & Insights     | customerInsightsModule | CRM, Party Reference Data Directory, Customer Financial Insights                           | Core (read model) | Read-only projection upstream client sync                     |
| Consumer Advisory Services           | advisoryServicesModule | Consumer Advisory Services, Customer Case Management                                       | Core              | Owns stage transition setup with outcome &                    |
| Advisory Proposal Construction       | proposalModule         | Consumer Advisory Services (proposal), Suitability Checking, Investment Portfolio Planning | Core              | Owns proposals RoA, versioning diffs                          |
| Regulatory & Suitability Compliance  | complianceModule       | Regulatory Compliance, Guideline Compliance, Suitability Checking                          | Core              | Sync validation only; async monitoring deferred               |
| Session Dialogue & Contact           | sessionDialogueModule  | Session Dialogue, Contact Handler, Servicing Event History                                 | Supporting        | Structure interaction capture 1 to cases                      |
| Document Services & Customer Consent | documentServicesModule | Document Services, Correspondence, Customer Consent, Customer Agreement                    | Supporting        | Document generation acceptance tracking, compliance artefacts |

| Bounded Context (BIAN-Aligned) | Module             | BIAN Service Domain(s)                         | Classification | Notes                         |
|--------------------------------|--------------------|------------------------------------------------|----------------|-------------------------------|
| Advisor Productivity           | productivityModule | (No BIAN equivalent – bank-specific extension) | Supporting     | Notebook, dictation notes lit |

## Deferred Contexts (not represented in current architecture)

| Context (BIAN-Aligned)                 | BIAN Service Domain(s)                           | Reason                              | Future Home                         |
|----------------------------------------|--------------------------------------------------|-------------------------------------|-------------------------------------|
| Party Authentication + Employee Access | Party Authentication, Employee Access            | Consumed from bank IAM (Entra ID)   | External dependency                 |
| Compliance Reporting                   | Compliance Reporting, Regulatory Reporting       | Vera's async mode not yet validated | Future module + event subscriptions |
| Servicing Activity Analysis (Gary)     | Servicing Activity Analysis, Employee Evaluation | Coach agent not in MVP scope        | Future module + agent               |
| Deal Execution & Fulfilment            | (Absorbed into Consumer Advisory Services)       | Terminal lifecycle stage            | Extract when complexity warrants    |

## BIAN Naming Convention

All bounded contexts and their corresponding modules use the following naming hierarchy:

1. **Primary BIAN service domain name** – used as the canonical label in API contracts, documentation, and integration specifications
2. **AIR qualifier** (where needed for disambiguation) – appended in parentheses, e.g. "Consumer Advisory Services (Lifecycle)" vs "Consumer Advisory Services (Proposal)"
3. **Bank-specific extension label** – used for contexts with no BIAN equivalent, clearly marked as extensions

## Where AIR Extends Beyond BIAN v14



| Extension                       | AIR Module                                     | Rationale                                                              |
|---------------------------------|------------------------------------------------|------------------------------------------------------------------------|
| AI Agent orchestration pattern  | Bob, Vera, Gary                                | BIAN models capabilities, not cross-cutting agent constructs           |
| Next-Best-Action prioritisation | Lead & Opportunity Management (domain service) | BIAN handles lead progression but not ranking algorithms               |
| Personal productivity tools     | Advisor Productivity                           | BIAN excludes internal employee productivity tooling                   |
| Event-driven backbone           | Cross-cutting                                  | BIAN models capabilities, not the event infrastructure connecting them |

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## Glossary

| Term     | Definition                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| PEP      | Policy Enforcement Point – the enterprise API gateway through which external service calls are routed                                         |
| AIR      | Advice & Intelligence Relationship – the platform being designed                                                                              |
| BIAN     | Banking Industry Architecture Network – industry-standard reference model for banking service domains                                         |
| Bob      | AI personal assistant agent – advisor's cross-context co-pilot (proposal drafting, queue reasoning, engagement prep, notebook interpretation) |
| Vera     | AI compliance agent – synchronous rule validation and suitability enforcement                                                                 |
| Gary     | AI professional coach agent – behavioural analysis and coaching insights (deferred)                                                           |
| RoA      | Record of Advice – regulatory documentation of advice given                                                                                   |
| FNA      | Financial Needs Analysis – client assessment inputs                                                                                           |
| HPA      | Horizontal Pod Autoscaler – Kubernetes auto-scaling mechanism                                                                                 |
| NBA      | Next-Best-Action – the prioritised recommendation for what an advisor should do next                                                          |
| QTD      | Quarter-to-Date – performance measurement period                                                                                              |
| GAP      | Pre-signature opportunity (not yet in pipeline)                                                                                               |
| Pipeline | Post-signature, pre-fulfilment opportunity (committed revenue)                                                                                |
| CRM      | Customer Relationship Management – BIAN service domain for managing customer relationships                                                    |

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*Date: May 2026*